

Agentic Computation Graph Corpus

Worked examples: one trace per extracted dataset · generated 2026-07-29

Every agent session in this corpus is normalised into one graph: nodes are runtime operations (LLM call, tool call, sub-agent boundary, user turn) and edges are data or control dependencies. This deck takes one trace from each extracted dataset and shows the whole thing — the graph, then every step's reasoning, tool arguments and result — so that what each source does and does not record is visible on the page rather than asserted in a table.

Design rule the corpus depends on: absent means null, never a guess. A field the source never shipped is printed below as "absent in this source".

dataset	domain	traces	nodes	med n	med dep	fan-out	reasoning	tokens
tracelab	coding	4,265	837,967	46	33	3/29	0%	43%
swe_rebench_openhands	coding	67,074	8,701,102	123	122	1/2	71%	0%
swe_agent_traj	coding	80,036	4,311,143	35	34	1/1	100%	0%
osworld_gelato	web	2,111	64,864	20	19	1/1	90%	0%

fan-out is median/max. reasoning = share of llm nodes with a stated rationale. tokens = share of all nodes with token accounting.

The four sources, and the trace shown for each

tracelab

Real developer sessions on their own machines. The only source with tokens, latency, timestamps and prefix-cache accounting — and the only one with named sub-agents. Sanitized, so it carries no text at all.
example: tracelab:claude:01aa3fdf-b691-c30e-2fbd-3b165f14cfe0 (35 nodes, depth 29)

swe_rebench_openhands

OpenHands agent (Qwen3-Coder-480B) on 1,823 real repositories. Structured tool calls plus the exact tool schemas the model was offered.
example: swe_rebench_openhands:chatcmpl-24d2b600e769398ddb64e7fb81cc3d8 (71 nodes, depth 70)

swe_agent_traj

SWE-agent scaffold (Llama-70b) on real repositories. Paired with OpenHands on purpose: same task type, different harness, so shape differences are attributable to the scaffold. Reasoning coverage is 100%.
example: swe_agent_traj:networkx__networkx-7024#769 (27 nodes, depth 26)

osworld_gelato

Agent driving real Chrome, LibreOffice, GIMP and VS Code in an Ubuntu VM via pyautogui. The only non-coding domain in the corpus.
example: osworld_gelato:gelato-30b:run_1:libreoffice_impress:5c1a6c3d-c1b3-47cb-9b01-8d1b7544ffa1 (22 nodes, depth 21)

Reading the graph pages

- nodes L = LLM call · T = tool call · A = sub-agent · U = user turn
- edges data (solid) = an observation feeds the next LLM call · control (dashed) = the LLM decided to call this tool
- agent_msg (green) = a human turn enters the round · order (dotted) = ordering only, no data passed
- columns topological level: nodes sharing a column have no dependency between them and could run concurrently

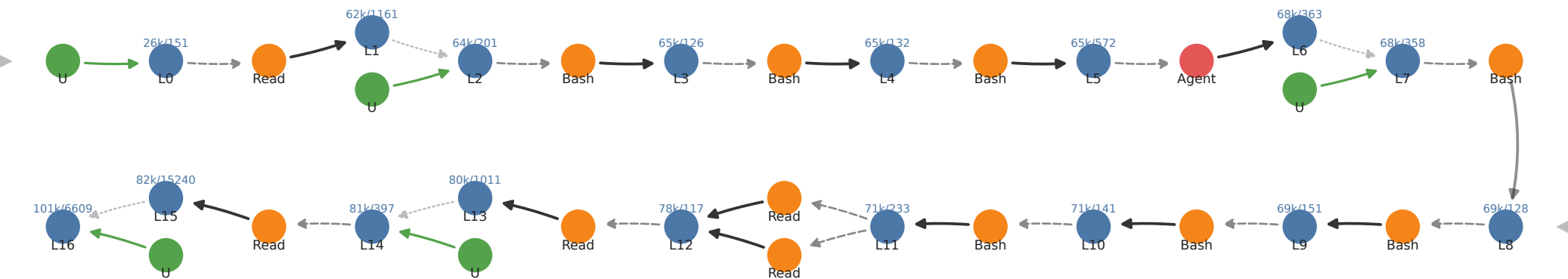
TraceLab — production Claude Code / Codex sessions

tracelab:claude:01aa3fdf-b691-c30e-2fbd-3b165f14cfe0 · record #2306 in tracelab/graphs.jsonl

Real developer sessions on their own machines. The only source with tokens, latency, timestamps and prefix-cache accounting — and the only one with named sub-agents. Sanitized, so it carries no text at all.

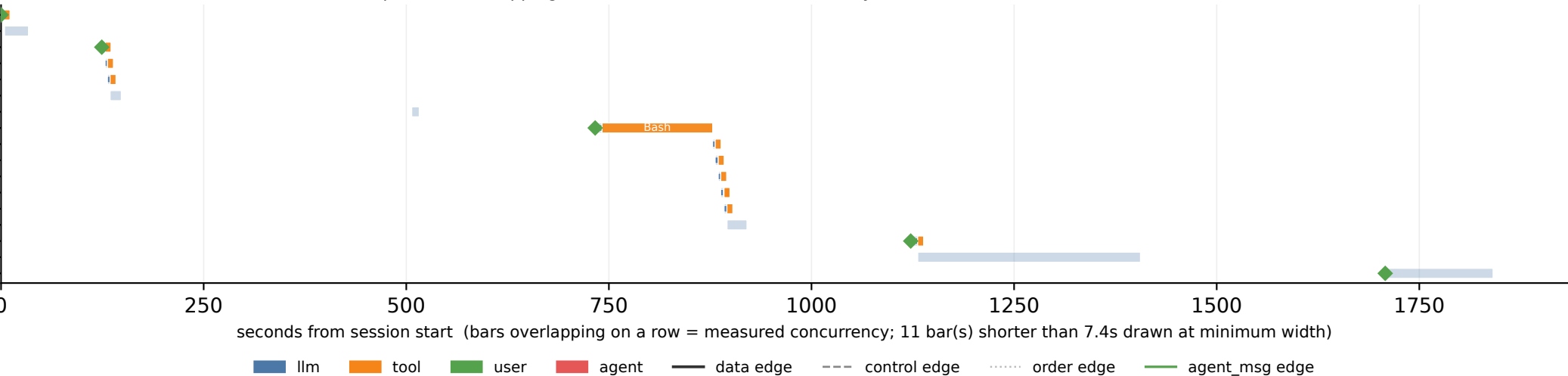
project=project_02893817 provider=claude n_rounds=17
models: claude-opus-4-6 x17
nodes 35 (llm 17, tool 12, agent 1, user 5) edges 35 (control 13, data 13, agent_msg 5, order 4)
depth 29 max fan-out 2 loop iterations 12 measured parallel width 2 DAG yes branches 1 (abandoned 0)
this trace — cost: tokens 49%, latency 71%, timestamps 100%, KV 49% semantic: reasoning 0% of llm, tool i/o 0% of actions
selection: drawn at random (seed 20260729) from 29 candidates: 8–40 nodes, ≥1 sub-agent node, among the first 4,265 records

Computation graph



topological level (siblings in one column may run concurrently) — wrapped every 15 levels; rows alternate direction (see ▶/◀)

Wall-clock execution timeline (real timestamps; bars overlapping on a row are measured concurrency)



TraceLab — production Claude Code / Codex sessions — step-by-step content

showing 6 of 17 steps · every field the trace holds for each node, verbatim and truncated only where marked · this source ships no system prompt and no task text

STEP 1 (turn_id 0) U0 → L0 → T0.0		
U0	USER	
edges	U0 →agent_msg→ L0	
cost	ts 23:26:58.100→23:26:58.100 · sizes: content 2,054 chars	
reasoning	– absent in this source (text stripped before release; only counts survive)	
L0	LLM	claude-opus-4-6
edges	U0 →agent_msg→ L0; L0 →control→ T0.0	
cost	in 26,738 tok (prefix-cache 24,918 = 93%) · out 151 tok · prefill 1,820 · wall 3,700 ms · ts 23:26:58.100→23:27:01.800	
reasoning	– absent in this source (text stripped before release; only counts survive)	
T0.0	TOOL	Read [ok]
edges	L0 →control→ T0.0; T0.0 →data→ L1	
cost	wall 602 ms · tool 602 ms · ts 23:27:01.800→23:27:02.402 · sizes: args 110 chars, result 60,945 chars	
input	– absent in this source (arguments stripped; 110 chars recorded)	
output	– absent in this source (result stripped; 60945 chars recorded)	
STEP 2 (turn_id 1) L1		
L1	LLM	claude-opus-4-6
edges	T0.0 →data→ L1; L1 →order→ L2	
cost	in 62,806 tok (prefix-cache 26,735 = 43%) · out 1,161 tok · prefill 36,071 · ts 23:27:02.402→23:27:31.901	
reasoning	– absent in this source (text stripped before release; only counts survive)	
STEP 3 (turn_id 2) U2 → L2 → T2.0		
U2	USER	
edges	U2 →agent_msg→ L2	
cost	ts 23:29:02.378→23:29:02.378 · sizes: content 254 chars	
reasoning	– absent in this source (text stripped before release; only counts survive)	
L2	LLM	claude-opus-4-6
edges	U2 →agent_msg→ L2; L1 →order→ L2; L2 →control→ T2.0	
cost	in 64,033 tok (prefix-cache 62,805 = 98%) · out 201 tok · prefill 1,228 · wall 4,100 ms · ts 23:29:02.378→23:29:06.478	
reasoning	– absent in this source (text stripped before release; only counts survive)	
T2.0	TOOL	Bash [ok]
edges	L2 →control→ T2.0; T2.0 →data→ L3	
cost	wall 95 ms · tool 95 ms · ts 23:29:06.478→23:29:06.573 · sizes: args 180 chars, result 2,686 chars	
input	– absent in this source (arguments stripped; 180 chars recorded)	
output	– absent in this source (result stripped; 2686 chars recorded)	
STEP 4 (turn_id 3) L3 → T3.0		
L3	LLM	claude-opus-4-6
edges	T2.0 →data→ L3; L3 →control→ T3.0	
cost	in 65,459 tok (prefix-cache 64,030 = 98%) · out 126 tok · prefill 1,429 · wall 2,789 ms · ts 23:29:06.573→23:29:09.362	
reasoning	– absent in this source (text stripped before release; only counts survive)	
T3.0	TOOL	Bash [ok]
edges	L3 →control→ T3.0; T3.0 →data→ L4	
cost	wall 81 ms · tool 81 ms · ts 23:29:09.362→23:29:09.443 · sizes: args 217 chars, result 131 chars	
input	– absent in this source (arguments stripped; 217 chars recorded)	
output	– absent in this source (result stripped; 131 chars recorded)	

TraceLab — production Claude Code / Codex sessions — step-by-step content (cont.)

showing 6 of 17 steps · every field the trace holds for each node, verbatim and truncated only where marked · this source ships no system prompt and no task text

STEP 5 (turn_id 4) L4 → T4.0

L4 **LLM** **claude-opus-4-6**
edges T3.0 →data→ L4; L4 →control→ T4.0
cost in 65,662 tok (prefix-cache 65,458 = 100%) · out 132 tok · prefill 204 · wall 3,128 ms · ts 23:29:09.443→23:29:12.571
reasoning — absent in this source (text stripped before release; only counts survive)

T4.0 **TOOL** **Bash** **[ok]**
edges L4 →control→ T4.0; T4.0 →data→ L5
cost wall 70 ms · tool 70 ms · ts 23:29:12.571→23:29:12.641 · sizes: args 233 chars, result 127 chars
input — absent in this source (arguments stripped; 233 chars recorded)
output — absent in this source (result stripped; 127 chars recorded)

STEP 6 (turn_id 5) L5 → A5.0

L5 **LLM** **claude-opus-4-6**
edges T4.0 →data→ L5; L5 →control→ A5.0
cost in 65,862 tok (prefix-cache 65,661 = 100%) · out 572 tok · prefill 201 · wall 13,808 ms · ts 23:29:12.641→23:29:26.449
reasoning — absent in this source (text stripped before release; only counts survive)

A5.0 **AGENT** **(SUB-AGENT)** **Agent**
edges L5 →control→ A5.0; A5.0 →data→ L6
cost wall 358,382 ms · tool 358,382 ms · ts 23:29:26.449→23:35:24.831 · sizes: args 1,606 chars, result 5,049 chars
input — absent in this source (arguments stripped; 1606 chars recorded)
output — absent in this source (result stripped; 5049 chars recorded)

... trace continues: 11 further steps not shown (whole-trace totals are on the graph page).

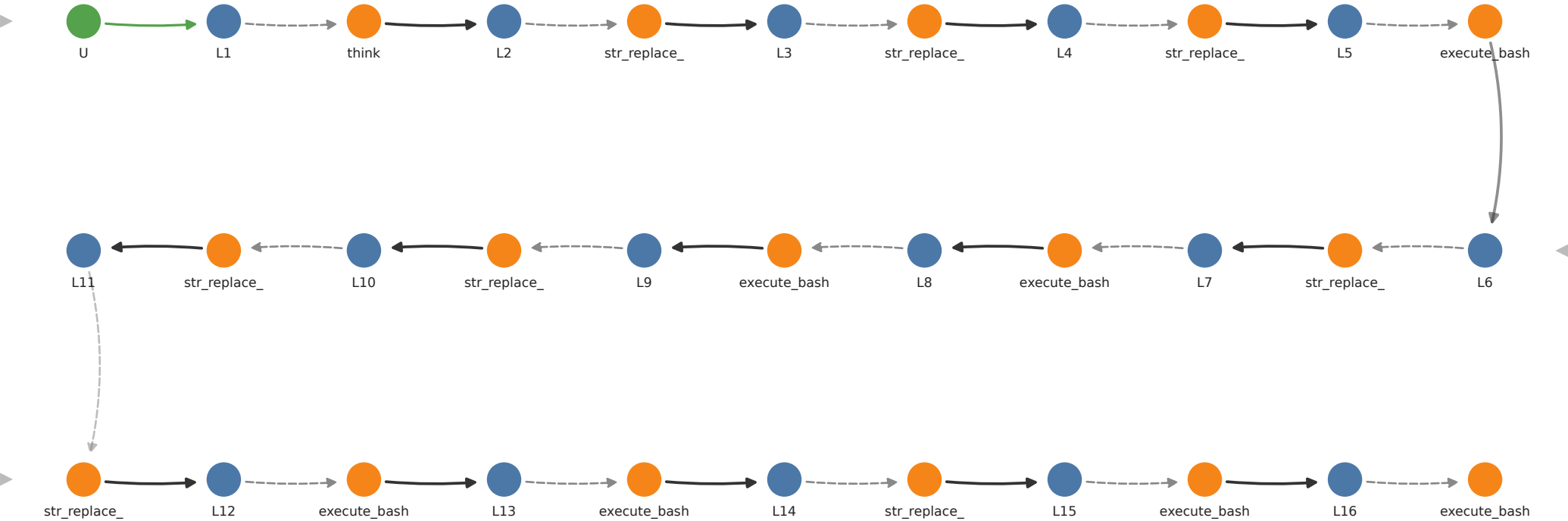
SWE-rebench / OpenHands — real GitHub issues

swe_rebench_openhands:chatcmpl-24d2b600e769398ddb64e7fb81cc3d8 · record #1090 in swe_rebench_openhands/graphs.jsonl

OpenHands agent (Qwen3-Coder-480B) on 1,823 real repositories. Structured tool calls plus the exact tool schemas the model was offered.

repo=astropy/photutils instance_id=astropy__photutils-1978 scaffold=openhands-v0.54.0 resolved=0 exit_status=submit gen_tests_correct=0.0
models: Qwen3-Coder-480B-A35B-Instruct x35
nodes 71 (llm 35, tool 35, user 1) edges 70 (control 35, data 34, agent_msg 1)
depth 70 max fan-out 1 loop iterations 34 measured parallel width - DAG yes branches 1 (abandoned 0)
this trace - cost: tokens 0%, latency 0%, timestamps 0%, KV 0% semantic: reasoning 77% of llm, tool i/o 100% of actions
selection: drawn at random (seed 20260729) from 67 candidates: 20-80 nodes, among the first 1,500 records

Computation graph — first 33 of 71 nodes (whole-trace figures above)



topological level (siblings in one column may run concurrently) — wrapped every 11 levels; rows alternate direction (see ▶/◀)

SWE-rebench / OpenHands — real GitHub issues — inputs to the run

what the model was given before step 1

SYSTEM PROMPT

```
sha256 9b89348423b86d94... 11,547 chars identical across trajectories, so it is stored once in data/graphs/swe_rebench_openhands/prompts.json
```

excerpt

You are OpenHands agent, a helpful AI assistant that can interact with a computer to solve tasks.

```
<ROLE>
Your primary role is to assist users by executing commands, modifying code, and solving technical problems effectively. You should be thorough, methodical, and prioritize quality over speed.
* If the user asks a question, like "why is X happening", don't try to fix the problem. Just give an answer to the question.
</ROLE>
```

```
<EFFICIENCY>
* Each action you take is somewhat expensive. Wherever possible, combine multiple actions into a single action, e.g. combine multiple bash commands into one, using sed and grep to edit/view multiple files at once.
* When exploring the codebase, use efficient tools like find, grep, and git commands with appropriate filters to minimize unnecessary operations.
... [+10.601 more characters in the trace]
```

TOOL SCHEMAS OFFERED	
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
19	19
20	20
21	21
22	22
23	23
24	24
25	25
26	26
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42	42
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44	44
45	45
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47	47
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69	69
70	70
71	71
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73	73
74	74
75	75
76	76
77	77
78	78
79	79
80	80
81	81
82	82
83	83
84	84
85	85
86	86
87	87
88	88
89	89
90	90
91	91
92	92
93	93
94	94
95	95
96	96
97	97
98	98
99	99
100	100

```
sha256 a4957103176dd6f6... 15.672 chars 5 tools: execute bash, think, finish, task tracker, str replace editor
```

excerpt

```
[{"function": {"description": "Execute a bash command in the terminal within a persistent shell session.\n\n\n### Command Execution\n* One command at a time: You can only execute one bash command at a time. If you need to run multiple commands sequentially, use `&&` or `;` to chain them together.\n* Persistent session: Commands execute in a persistent shell session where environment variables, virtual environments, and working directory persist between commands.\n* Soft timeout: Commands have a soft timeout of 10 seconds, once that's reached, you have the option to continue or interrupt the command (see section below for details)\n\n\n### Long-running Commands\n* For commands that may run indefinitely, run them in the background and redirect output to a file, e.g. `python3 app.py > server.log 2>&1 &`. \n* For commands that may run for a long time (e.g. installation or testing commands), or commands that run for a fixed amount of time (e.g. sleep), you should set the `\"timeout\"` parameter of your function call to an appropriate value.\n* If a bash command returns exit code `-1`, this means the process hit the soft timeout and is not yet finished. By setting `is_input` to `true`, you can:\n    - Send empty `command` to retrieve additional logs\n    - Send text (set `command` to the text) to STDOUT of the running process\n    - Send command to kill the process\n\n\n[+14,328 more characters in the trace]"}]
```

TASK / INPUT PROMPT	
1	Write a short story about a person who discovers a hidden talent.
2	Generate a list of 10 creative ideas for a science fiction movie.
3	Compose a poem about the beauty of nature.
4	Describe the process of photosynthesis in simple terms.
5	Explain the importance of recycling to a young child.
6	Write a persuasive essay about the benefits of renewable energy.
7	Summarize the main points of a research paper on climate change.
8	Translate the sentence 'The cat sat on the mat' into Spanish.
9	Identify the main character and theme of the novel 'Pride and Prejudice'.
10	Calculate the area of a circle with a radius of 5 units.

```
1 user node(s); first shown. sizes: content 5,474 chars
```

text

```
<uploaded_files>
/workspace/astropy__photutils_2.0
</uploaded_files>
```

I've uploaded a python code repository in the directory `astropy_photutils_2.0`. Consider the following issue description:

`<issue_description>`
Sources that match `sharplo(hi)` and `roundlo(hi)` limits should be included
Currently star finding code filters out sources that have sharpness or roundness match one of the limits for these parameter, see <https://github.com/astropy/photutils/blob/c15b004be4ecd1256f8a8db8c9c7a52e795fe3a8/photutils/detection/irafstarfinder.py#L580-L583>

I believe the strict inequalities in the code above should be replaced with `>=` and `<=`.

Here is an example of a source that should be detected:

```
```python
In [1]: import numpy as np

In [2]: from photutils import detection
```

... [+4,578 more characters in the trace]

# SWE-rebench / OpenHands — real GitHub issues — step-by-step content

showing 6 of 35 steps · every field the trace holds for each node, verbatim and truncated only where marked

STEP 1 (turn_id 1) L1 → T1.0	
L1	LLM Qwen3-Coder-480B-A35B-Instruct
edges	U0 -agent_msg→ L1; L1 -control→ T1.0
cost	sizes: reasoning 286 chars
reasoning	I'll help you implement the necessary changes to fix the issue with star finding code filtering out sources that match sharpness or roundness limits. Let me start by understanding the problem and working through each phase systematically.   ## Phase 1. READING: Understanding the Problem
T1.0	TOOL think [unknown]
edges	L1 -control→ T1.0; T1.0 -data→ L2
cost	sizes: args 1,576 chars, result 29 chars
input	<pre>{"thought": "Let me carefully read and understand the issue description:\n\n1. The problem is in the IRAFStarFinder class in photutils/detection/irafstarfinder.py\n2. Currently, the code uses strict inequalities (&lt; and &gt;) when filtering sources based on sharpness and roundness parameters\n3. This means sources that exactly match the limit values (sharplo, sharphi, roundlo, roundhi) are incorrectly filtered out\n4. The issue shows that perfectly round stars (roundness = 0.0) are excluded when roundlo=0 because the condition uses \"&gt; roundlo\" instead of \"&gt;= roundlo\"\n5. The fix should change strict inequalities to inclusive inequalities (&gt;= and &lt;=)\n\nKey technical details:\n- File: photutils/detection/irafstarfinder.py, lines 580-583\n- Parameters involved: sharplo, sharphi, roundlo, roundhi\n- Current logic: strict inequalities exclude boundary values\n- Expected logic: inclusive inequalities should include boundary values\n- Example shows roundness=0.0 being excluded when roundlo=0, but included when roundlo=-1e-12\n\nSteps to reproduce:\n1. Create a test image with a star-like pattern\n2. Use IRAFStarFinder with roundlo=0\n3. Ob</pre>  ... [+424 more characters in the trace]
output	Your thought has been logged.
STEP 2 (turn_id 2) L2 → T2.0	
L2	LLM Qwen3-Coder-480B-A35B-Instruct
edges	T1.0 -data→ L2; L2 -control→ T2.0
cost	sizes: reasoning 1,280 chars
reasoning	### 1.1-1.3 Problem Analysis   The issue is in the <b>IRAFStarFinder</b> class in the <b>photutils/detection/irafstarfinder.py</b> file. The problem involves incorrect filtering logic for <b>sharpness</b> and <b>roundness</b> parameters:      - <b>Current behavior</b>: Uses strict inequalities (<code>&gt;</code> and <code>&lt;</code>) which exclude sources that exactly match the boundary values    - <b>Expected behavior</b>: Should use inclusive inequalities (<code>&gt;=</code> and <code>&lt;=</code>) to include sources that match the boundary limits      ... [+786 more characters in the trace]
T2.0	TOOL str_replace_editor [unknown]
edges	L2 -control→ T2.0; T2.0 -data→ L3
cost	sizes: args 41 chars, result 756 chars
input	<pre>{"command": "view", "path": "/workspace"}</pre>
output	Here's the files and directories up to 2 levels deep in /workspace, excluding hidden items:  <pre>/workspace/ /workspace/astropy__photutils__2.0/ /workspace/astropy__photutils__2.0/=0.11 /workspace/astropy__photutils__2.0/CHANGES.rst /workspace/astropy__photutils__2.0/CITATION.md /workspace/astropy__photutils__2.0/CODE_OF_CONDUCT.rst</pre>  ... [+418 more characters in the trace]

# SWE-rebench / OpenHands — real GitHub issues — step-by-step content (cont.)

showing 6 of 35 steps · every field the trace holds for each node, verbatim and truncated only where marked

STEP 3 (turn_id 3) L3 → T3.0	
L3	LLM Qwen3-Coder-480B-A35B-Instruct
edges	T2.0 →data→ L3; L3 →control→ T3.0
cost	no tokens, latency or timestamps – absent in this source
reasoning	– absent in this source for this node – the model acted without narrating
T3.0	T00L str_replace_editor [unknown]
edges	L3 →control→ T3.0; T3.0 →data→ L4
cost	sizes: args 65 chars, result 3,075 chars
input	
output	<pre>{"path": "/workspace/astropy_photutils__2.0", "command": "view"}</pre>  Here's the files and directories up to 2 levels deep in /workspace/astropy_photutils__2.0, excluding hidden items:  <pre>/workspace/astropy_photutils__2.0/ /workspace/astropy_photutils__2.0/=0.11 /workspace/astropy_photutils__2.0/CHANGES.rst /workspace/astropy_photutils__2.0/CITATION.md /workspace/astropy_photutils__2.0/CODE_OF_CONDUCT.rst /workspace/astropy_photutils__2.0/CONTRIBUTING.rst ... [+2,634 more characters in the trace]</pre>
STEP 4 (turn_id 4) L4 → T4.0	
L4	LLM Qwen3-Coder-480B-A35B-Instruct
edges	T3.0 →data→ L4; L4 →control→ T4.0
cost	no tokens, latency or timestamps – absent in this source
reasoning	– absent in this source for this node – the model acted without narrating
T4.0	T00L str_replace_editor [unknown]
edges	L4 →control→ T4.0; T4.0 →data→ L5
cost	sizes: args 76 chars, result 4,260 chars
input	
output	<pre>{"path": "/workspace/astropy_photutils__2.0/README.rst", "command": "view"}</pre>  Here's the result of running `cat -n` on /workspace/astropy_photutils__2.0/README.rst:  <pre>1  ===== 2  Photutils 3  ===== 4 5   PyPI Version   Conda Version   PyPI Downloads   Astropy  6 ... [+4,210 more characters in the trace]</pre>
STEP 5 (turn_id 5) L5 → T5.0	
L5	LLM Qwen3-Coder-480B-A35B-Instruct
edges	T4.0 →data→ L5; L5 →control→ T5.0
cost	sizes: reasoning 65 chars
reasoning	Let me check if there are any specific test running instructions:
T5.0	T00L execute_bash [unknown]
edges	L5 →control→ T5.0; T5.0 →data→ L6
cost	sizes: args 156 chars, result 407 chars
input	
output	<pre>{"command": "cd /workspace/astropy_photutils__2.0 &amp;&amp; find . -name \"*.md\" -o -name \"*.rst\" -o -name \"tox.ini\" -o -name \"pyproject.toml\"   head -10"}</pre>



# SWE-rebench / OpenHands — real GitHub issues — step-by-step content (cont.)

showing 6 of 35 steps · every field the trace holds for each node, verbatim and truncated only where marked

output

```
./LICENSE.rst
./tox.ini
./README.rst
./CONTRIBUTING.rst
./CODE_OF_CONDUCT.rst
./CITATION.md
./CHANGES.rst
... [+295 more characters in the trace]
```

STEP 6 (turn\_id 6) L6 → T6.0

L6 LLM Qwen3-Coder-480B-A35B-Instruct

```
edges T5.0 -data→ L6; L6 -control→ T6.0
cost no tokens, latency or timestamps – absent in this source
reasoning – absent in this source for this node – the model acted without narrating
```

T6.0 TOOL str\_replace\_editor [unknown]

```
edges L6 -control→ T6.0; T6.0 -data→ L7
cost sizes: args 80 chars, result 8,499 chars
```

input

```
{"path": "/workspace/astropy__photutils__2.0/pyproject.toml", "command": "view"}
```

output

```
Here's the result of running `cat -n` on /workspace/astropy__photutils__2.0/pyproject.toml:
1 [project]
2 name = 'photutils'
3 description = 'An Astropy package for source detection and photometry'
4 readme = 'README.rst'
5 license = {file = 'LICENSE.rst'}
6 authors = [
... [+8,806 more characters in the trace]
```

... trace continues: 29 further steps not shown (whole-trace totals are on the graph page).

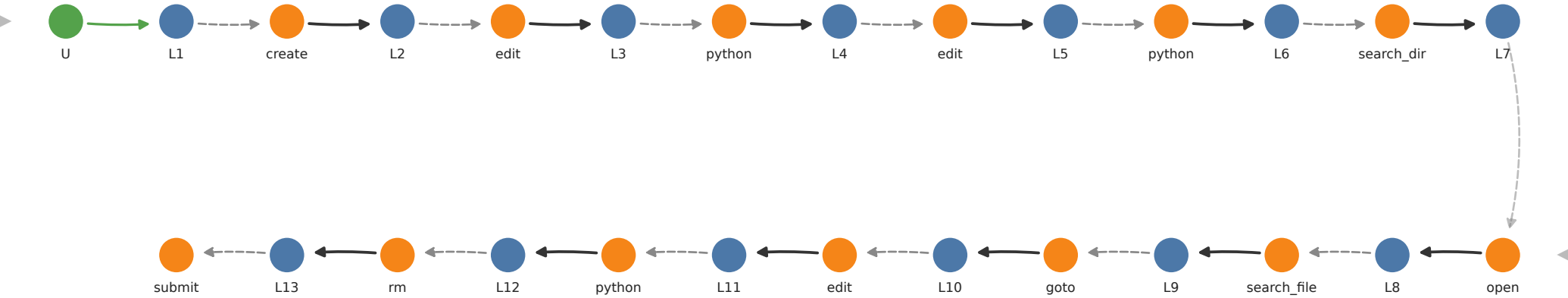
# SWE-agent — same domain, different harness

swe\_agent\_traj:networkx\_\_networkx-7024#769 · record #769 in swe\_agent\_traj/graphs.jsonl

SWE-agent scaffold (Llama-70b) on real repositories. Paired with OpenHands on purpose: same task type, different harness, so shape differences are attributable to the scaffold. Reasoning coverage is 100%.

instance\_id=networkx\_\_networkx-7024 model\_name=swe-agent-llama-70b exit\_status=submitted  
models: swe-agent-llama-70b x13  
nodes 27 (llm 13, tool 13, user 1) edges 26 (control 13, data 12, agent\_msg 1)  
depth 26 max fan-out 1 loop iterations 12 measured parallel width — DAG yes branches 1 (abandoned 0)  
this trace — cost: tokens 0%, latency 0%, timestamps 0%, KV 0% semantic: reasoning 100% of llm, tool i/o 100% of actions  
selection: drawn at random (seed 20260729) from 1,592 candidates: 16–34 nodes, among the first 4,000 records

Computation graph



topological level (siblings in one column may run concurrently) — wrapped every 14 levels; rows alternate direction (see ►/◄)

■ llm   ■ tool   ■ user   — data edge   - - - control edge   — agent\_msg edge

# SWE-agent — same domain, different harness — inputs to the run

what the model was given before step 1

## SYSTEM PROMPT

sha256 cbb6b0428d8d287e... 4,873 chars identical across trajectories, so it is stored once in data/graphs/swe\_agent\_traj/prompts.json

### excerpt

SETTING: You are an autonomous programmer, and you're working directly in the command line with a special interface.

The special interface consists of a file editor that shows you 100 lines of a file at a time.

In addition to typical bash commands, you can also use the following commands to help you navigate and edit files.

#### COMMANDS:

##### open:

docstring: opens the file at the given path in the editor. If line\_number is provided, the window will be move to include that line

signature: open <path> [<line\_number>]

##### arguments:

- path (string) [required]: the path to the file to open

... [+4,197 more characters in the trace]

## TOOL SCHEMAS OFFERED

schemas — absent in this source — the offered action space is not recoverable

## TASK / INPUT PROMPT

1 user node(s); first shown. sizes: content 4,255 chars

### text

We're currently solving the following issue within our repository. Here's the issue text:

#### ISSUE:

```is\_k\_edge\_connected``` incorrectly returns True for k=2 with multi-component graphs without bridges

Current Behavior

The implementation of ```is_k_edge_connected``` currently defers to ```is_connected``` for k=1, to ```has_bridges``` for k=2, and runs the full check for k>2.

As a result, this returns True for a multi-component graph without bridges.

Expected Behavior

Since the graph has multiple components, it is not connected in the first place and ```is_k_edge_connected``` should return False for all (positive) values of k.

Steps to Reproduce

```

import networkx as nx

G = nx.compose(nx.complete\_graph([0, 1, 2]), nx.complete\_graph([3, 4, 5]))

nx.connectivity.is\_k\_edge\_connected(G, 2)

... [+3,362 more characters in the trace]

# SWE-agent — same domain, different harness — step-by-step content

showing 6 of 13 steps · every field the trace holds for each node, verbatim and truncated only where marked

| STEP 1 (turn_id 1) L1 → T1.0 |                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| L1                           | LLM                                                                                                                                                                                                                                                                                                                                                                                                                      | swe-agent-llama-70b |
| edges                        | U0 →agent_msg→ L1; L1 →control→ T1.0                                                                                                                                                                                                                                                                                                                                                                                     |                     |
| cost                         | sizes: reasoning 232 chars                                                                                                                                                                                                                                                                                                                                                                                               |                     |
| reasoning                    | To start addressing the issue, we need to reproduce the problem as described. The issue includes some example code for reproduction, which we can use. We'll create a new file called `reproduce.py` and paste the example code into it.                                                                                                                                                                                 |                     |
| T1.0                         | TOOL                                                                                                                                                                                                                                                                                                                                                                                                                     | create [unknown]    |
| edges                        | L1 →control→ T1.0; T1.0 →data→ L2                                                                                                                                                                                                                                                                                                                                                                                        |                     |
| cost                         | sizes: command 19 chars, result 124 chars                                                                                                                                                                                                                                                                                                                                                                                |                     |
| input                        | create reproduce.py                                                                                                                                                                                                                                                                                                                                                                                                      |                     |
| output                       | [File: /networkx/reproduce.py (1 lines total)]<br>1:<br><br>(Open file: /networkx/reproduce.py)<br>(Current directory: /networkx)<br>bash-\$                                                                                                                                                                                                                                                                             |                     |
| STEP 2 (turn_id 2) L2 → T2.0 |                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |
| L2                           | LLM                                                                                                                                                                                                                                                                                                                                                                                                                      | swe-agent-llama-70b |
| edges                        | T1.0 →data→ L2; L2 →control→ T2.0                                                                                                                                                                                                                                                                                                                                                                                        |                     |
| cost                         | sizes: reasoning 76 chars                                                                                                                                                                                                                                                                                                                                                                                                |                     |
| reasoning                    | Now let's paste in the example code from the issue to reproduce the problem.                                                                                                                                                                                                                                                                                                                                             |                     |
| T2.0                         | TOOL                                                                                                                                                                                                                                                                                                                                                                                                                     | edit [unknown]      |
| edges                        | L2 →control→ T2.0; T2.0 →data→ L3                                                                                                                                                                                                                                                                                                                                                                                        |                     |
| cost                         | sizes: command 160 chars, result 422 chars                                                                                                                                                                                                                                                                                                                                                                               |                     |
| input                        | edit 1:1<br>import networkx as nx<br><br>G = nx.compose(nx.complete_graph([0, 1, 2]), nx.complete_graph([3, 4, 5]))<br>nx.connectivity.is_k_edge_connected(G, 2)<br>end_of_edit                                                                                                                                                                                                                                          |                     |
| output                       | [File: /networkx/reproduce.py (4 lines total)]<br>1:import networkx as nx<br>2:<br>3:G = nx.compose(nx.complete_graph([0, 1, 2]), nx.complete_graph([3, 4, 5]))<br>4:nx.connectivity.is_k_edge_connected(G, 2)<br>File updated. Please review the changes and make sure they are correct (correct indentation, no duplicate lines, etc). Edit the file again if necessary.<br><br>... [+71 more characters in the trace] |                     |

# SWE-agent — same domain, different harness — step-by-step content (cont.)

showing 6 of 13 steps · every field the trace holds for each node, verbatim and truncated only where marked

STEP 3 (turn\_id 3) L3 → T3.0

L3 LLM swe-agent-llama-70b

edges T2.0 →data→ L3; L3 →control→ T3.0

cost sizes: reasoning 121 chars

reasoning

The example code has been added to the `reproduce.py` file. Now, let's run the code to see if we can reproduce the issue.

T3.0 TOOL python [unknown]

edges L3 →control→ T3.0; T3.0 →data→ L4

cost sizes: command 19 chars, result 239 chars

input

python reproduce.py

output

/networkx/networkx/utils/backends.py:135: RuntimeWarning: networkx backend defined more than once: nx-loopback  
backends.update(\_get\_backends("networkx.backends"))  
  
(Open file: /networkx/reproduce.py)  
(Current directory: /networkx)  
bash-\$

STEP 4 (turn\_id 4) L4 → T4.0

L4 LLM swe-agent-llama-70b

edges T3.0 →data→ L4; L4 →control→ T4.0

cost sizes: reasoning 407 chars

reasoning

It looks like there's a warning about the networkx backend being defined more than once, but this doesn't affect our ability to reproduce the issue. The warning is noted, but we can proceed.  
  
Since the `is\_k\_edge\_connected` function returns a boolean value, we should print the result to confirm whether the function is returning the expected value. Let's modify the `reproduce.py` file to print the result.

T4.0 TOOL edit [unknown]

edges L4 →control→ T4.0; T4.0 →data→ L5

cost sizes: command 69 chars, result 429 chars

input

edit 4:4  
print(nx.connectivity.is\_k\_edge\_connected(G, 2))  
end\_of\_edit

output

[File: /networkx/reproduce.py (4 lines total)]  
1:import networkx as nx  
2:  
3:G = nx.compose(nx.complete\_graph([0, 1, 2]), nx.complete\_graph([3, 4, 5]))  
4:print(nx.connectivity.is\_k\_edge\_connected(G, 2))  
File updated. Please review the changes and make sure they are correct (correct indentation, no duplicate lines, etc). Edit the file again if necessary.  
  
... [+71 more characters in the trace]

# SWE-agent — same domain, different harness — step-by-step content (cont.)

showing 6 of 13 steps · every field the trace holds for each node, verbatim and truncated only where marked

STEP 5 (turn\_id 5) L5 → T5.0

L5 LLM swe-agent-llama-70b

edges T4.0 -data→ L5; L5 -control→ T5.0

cost sizes: reasoning 145 chars

reasoning

The `reproduce.py` file has been updated to print the result of the `is\_k\_edge\_connected` function. Let's run the script again to see the output.

T5.0 TOOL python [unknown]

edges L5 -control→ T5.0; T5.0 -data→ L6

cost sizes: command 19 chars, result 244 chars

input

python reproduce.py

output

/networkx/networkx/utils/backends.py:135: RuntimeWarning: networkx backend defined more than once: nx-loopback  
backends.update(\_get\_backends("networkx.backends"))  
True  
  
(Open file: /networkx/reproduce.py)  
(Current directory: /networkx)  
bash-\$

STEP 6 (turn\_id 6) L6 → T6.0

L6 LLM swe-agent-llama-70b

edges T5.0 -data→ L6; L6 -control→ T6.0

cost sizes: reasoning 489 chars

reasoning

The output of the `is\_k\_edge\_connected` function is `True`, which confirms the issue. The function should return `False` for a multi-component graph without bridges.

The next step is to locate the implementation of the `is\_k\_edge\_connected` function and understand why it's returning `True` for k=2 with multi-component graphs without bridges. We need to find the file and line number where this function is defined. Let's search for the `is\_k\_edge\_connected` function in the repository.

T6.0 TOOL search\_dir [unknown]

edges L6 -control→ T6.0; T6.0 -data→ L7

cost sizes: command 36 chars, result 268 chars

input

search\_dir "def is\_k\_edge\_connected"

output

Found 1 matches for "def is\_k\_edge\_connected" in /networkx:  
/networkx/networkx/algorithms/connectivity/edge\_augmentation.py (1 matches)  
End of matches for "def is\_k\_edge\_connected" in /networkx  
  
(Open file: /networkx/reproduce.py)  
(Current directory: /networkx)  
bash-\$

... trace continues: 7 further steps not shown (whole-trace totals are on the graph page).

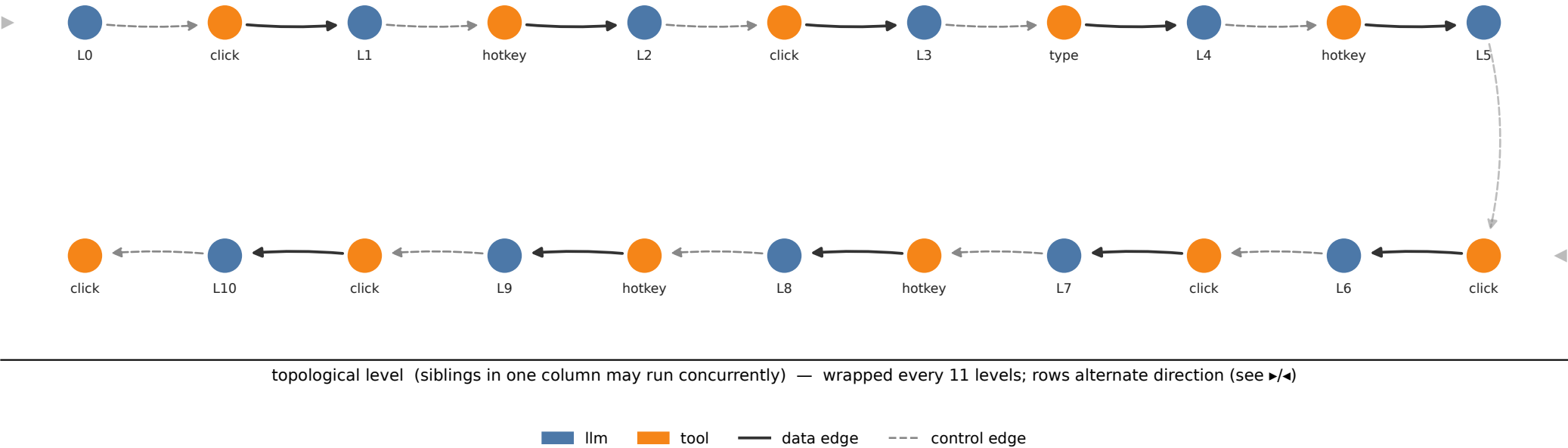
# OSWorld / Gelato — GUI computer-use on real desktop apps

osworld\_gelato:gelato-30b:run\_1:libreoffice\_impress:5c1a6c3d-c1b3-47cb-9b01-8d1b7544ffa1 · record #131 in osworld\_gelato/graphs.jsonl

Agent driving real Chrome, LibreOffice, GIMP and VS Code in an Ubuntu VM via pyautogui. The only non-coding domain in the corpus.

application=libreoffice\_impress task\_id=5c1a6c3d-c1b3-47cb-9b01-8d1b7544ffa1 steps=11 result\_score=1.0  
models: gelato-30b x11  
nodes 22 (llm 11, tool 11) edges 21 (control 11, data 10)  
depth 21 max fan-out 1 loop iterations 10 measured parallel width - DAG yes branches 1 (abandoned 0)  
this trace - cost: tokens 0%, latency 0%, timestamps 0%, KV 0% semantic: reasoning 100% of llm, tool i/o 100% of actions  
selection: drawn at random (seed 20260729) from 623 candidates: 16-34 nodes, among the first 2,111 records

Computation graph



# OSWorld / Gelato — GUI computer-use on real desktop apps — step-by-step content

showing 5 of 11 steps · every field the trace holds for each node, verbatim and truncated only where marked · this source ships no system prompt and no task text

| STEP 1 (turn_id 0) L0 → T0.0 |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| L0                           | LLM                                                                                                                                                                                                                                                                                                                                                                                                                                                  | gelato-30b       |
| edges                        | L0 →control→ T0.0                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
| cost                         | sizes: reasoning 465 chars                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |
| reasoning                    | <div>Checklist:</div> <ul style="list-style-type: none"><li>- Confirm slide 1 is active and locate the title textbox.</li><li>- Enter the title textbox to enable text editing.</li><li>- Bold the title text, set font size to 44pt, and underline it.</li><li>- Look for any other editable text on slide 1 and bold it.</li><li>- Review slide 1 to ensure all requirements are met.</li></ul> <div>... [+169 more characters in the trace]</div> |                  |
| T0.0                         | T00L                                                                                                                                                                                                                                                                                                                                                                                                                                                 | click [unknown]  |
| edges                        | L0 →control→ T0.0; T0.0 →data→ L1                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
| cost                         | sizes: args 255 chars, command 87 chars, result 114 chars                                                                                                                                                                                                                                                                                                                                                                                            |                  |
| input                        | {<br>"instruction": "Double-click directly on the white text reading “Business Infographic Template” centered on the left blue area of slide 1 to place the text cursor inside that textbox for editing.",<br>"num_clicks": 2,<br>"button_type": "left",<br>"hold_keys": []<br>}                                                                                                                                                                     |                  |
| executed                     | import pyautogui; import pyautogui; pyautogui.click(463, 529, clicks=2, button='left');                                                                                                                                                                                                                                                                                                                                                              |                  |
| output                       | Action (tool call) is executed. For your reference, you have maximum of 50 steps, and current step is 1 out of 50.                                                                                                                                                                                                                                                                                                                                   |                  |
| STEP 2 (turn_id 1) L1 → T1.0 |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |
| L1                           | LLM                                                                                                                                                                                                                                                                                                                                                                                                                                                  | gelato-30b       |
| edges                        | T0.0 →data→ L1; L1 →control→ T1.0                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
| cost                         | sizes: reasoning 127 chars                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |
| reasoning                    | Purpose: Select all text in the active title textbox so formatting will apply to the entire title. Minimal input: press Ctrl+A.                                                                                                                                                                                                                                                                                                                      |                  |
| T1.0                         | T00L                                                                                                                                                                                                                                                                                                                                                                                                                                                 | hotkey [unknown] |
| edges                        | L1 →control→ T1.0; T1.0 →data→ L2                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
| cost                         | sizes: args 23 chars, command 47 chars, result 114 chars                                                                                                                                                                                                                                                                                                                                                                                             |                  |
| input                        | {<br>"keys": ["ctrl", "a"]<br>}                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |
| executed                     | import pyautogui; pyautogui.hotkey('ctrl', 'a')                                                                                                                                                                                                                                                                                                                                                                                                      |                  |
| output                       | Action (tool call) is executed. For your reference, you have maximum of 50 steps, and current step is 2 out of 50.                                                                                                                                                                                                                                                                                                                                   |                  |
| STEP 3 (turn_id 2) L2 → T2.0 |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |
| L2                           | LLM                                                                                                                                                                                                                                                                                                                                                                                                                                                  | gelato-30b       |
| edges                        | T1.0 →data→ L2; L2 →control→ T2.0                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
| cost                         | sizes: reasoning 151 chars                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |
| reasoning                    | Purpose: Focus the font size field to change the title size to 44 pt. Minimal input: click on the “20 pt” size box in the Character panel on the right.                                                                                                                                                                                                                                                                                              |                  |
| T2.0                         | T00L                                                                                                                                                                                                                                                                                                                                                                                                                                                 | click [unknown]  |
| edges                        | L2 →control→ T2.0; T2.0 →data→ L3                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |
| cost                         | sizes: args 189 chars, command 88 chars, result 114 chars                                                                                                                                                                                                                                                                                                                                                                                            |                  |
| input                        | {<br>"instruction": "Click the font size field showing “20 pt” in the right-side Properties > Character panel to focus it for editing.",<br>"num_clicks": 1,<br>"button_type": "left",<br>"hold_keys": []<br>}                                                                                                                                                                                                                                       |                  |



# OSWorld / Gelato — GUI computer-use on real desktop apps — step-by-step content (cont.)

showing 5 of 11 steps · every field the trace holds for each node, verbatim and truncated only where marked · this source ships no system prompt and no task text

executed

import pyautogui; import pyautogui; pyautogui.click(1786, 237, clicks=1, button='left');

output

Action (tool call) is executed. For your reference, you have maximum of 50 steps, and current step is 3 out of 50.

STEP 4 (turn\_id 3) L3 → T3.0

L3 LLM gelato-30b

edges T2.0 →data→ L3; L3 →control→ T3.0

cost sizes: reasoning 124 chars

reasoning

Purpose: Set the selected title text to 44 pt size. Minimal input: type 44 into the focused font size field and press Enter.

T3.0 TOOL type [unknown]

edges L3 →control→ T3.0; T3.0 →data→ L4

cost sizes: args 187 chars, command 155 chars, result 114 chars

input

{"element\_description": "The font size input field in the right-side Properties > Character panel, currently focused and showing “20 pt”.", "text": "44", "overwrite": true, "enter": true}

executed

import pyautogui; pyautogui.click(1786, 237); pyautogui.hotkey('ctrl', 'a'); pyautogui.press('backspace'); pyautogui.write('44'); pyautogui.press('enter');

output

Action (tool call) is executed. For your reference, you have maximum of 50 steps, and current step is 4 out of 50.

STEP 5 (turn\_id 4) L4 → T4.0

L4 LLM gelato-30b

edges T3.0 →data→ L4; L4 →control→ T4.0

cost sizes: reasoning 103 chars

reasoning

Purpose: Underline the selected title text to meet formatting requirement. Minimal input: press Ctrl+U.

T4.0 TOOL hotkey [unknown]

edges L4 →control→ T4.0; T4.0 →data→ L5

cost sizes: args 23 chars, command 47 chars, result 114 chars

input

{"keys": ["ctrl", "u"]}

executed

import pyautogui; pyautogui.hotkey('ctrl', 'u')

output

Action (tool call) is executed. For your reference, you have maximum of 50 steps, and current step is 5 out of 50.

⋮ trace continues: 6 further steps not shown (whole-trace totals are on the graph page).

# What the four examples show together

each claim below is visible on the preceding pages

---

## 1. Cost and semantics are disjoint in public data.

TraceLab pages carry tokens, prefix-cache splits, latency and timestamps on every step and not one word of text. The other three carry reasoning, arguments and observations verbatim and zero cost fields (43% vs 0% token coverage). Relating why a node ran to what it cost needs two datasets, or an instrumented run of our own.

## 2. The harness decides graph shape, not the model.

Across 13,915,076 nodes, the three benchmark sources never exceed fan-out 2 — every step is one tool, so the graph is the chain. TraceLab's production sessions reach fan-out 29 with a median of 3, and 88.2% of its multi-tool rounds overlap in measured wall-clock time. A corpus built only from benchmark rollouts would conclude, wrongly, that agent computation graphs are inherently serial.

## 3. Trace size is a configuration setting.

OpenHands stops at 201 nodes (100 iterations) and OSWorld at 100 nodes (50 steps); SWE-agent has no cap and decays smoothly to 817. TraceLab is unbounded: p99 2,722, max 18,482.

## 4. Only production traffic has a conversation in it.

The benchmark traces each begin with exactly one injected task statement and then run uninterrupted until a cap or a `submit`. TraceLab sessions have a human interrupting, redirecting and resuming — mean 11.3 user turns, and 4.2% of sessions span more than a day. That is what produces prompt-cache reuse across hours, which a benchmark rollout never exercises.

## 5. Sub-agents appear in production and nowhere else.

1,867 `agent` nodes across 300 of 4,265 TraceLab sessions (Claude `Agent`, Codex `spawn\_agent`/`wait\_agent`); all three semantic sources are single-agent. No dataset in hand has reasoning AND sub-agents together, which is the clearest gap to target next.

---

Reproduce this deck: `.venv/bin/python scripts/render_showcase_pdf.py --seed <n>`. Traces are sampled from a seeded, legibility-constrained pool; the constraint is printed on each dataset's graph page.